

EXPLORATORY DATA ANALYSIS — POST-INGESTION ADDENDUM

Dataset 2: Top 50 — 73 Countries (2023–2025)
Project: Capstone | Course: DATA206 | Student: Leena Komenski
Ingestion completed: April 20, 2026

This addendum confirms the post-ingestion validation results for Dataset 2. All findings below were obtained via SQL Server query after BULK INSERT of Dataset 2 into the HiddenGemMusic database.

9. Post-Ingestion Validation Results

9.1 Final Row Counts — Confirmed

Metric	Pre-Ingestion Value	Post-Ingestion Actual	Status
Total rows after dedup	2,110,316	2,110,316	✓ Exact match — no duplicates removed
Earliest snapshot_date	Oct 18 2023	Oct 18 2023	✓ Confirmed
Latest snapshot_date	Jun 11 2025	Jun 11 2025	✓ Confirmed
Global rows (iso_code '--')	28,908	28,908	✓ Exact match
NULL trend_velocity rows	Expected 0	0	✓ All rows normalised correctly

9.2 Movement Column Finding — Confirmed

The zero NULL finding for daily_movement and weekly_movement documented in Section 5 of the original EDA was confirmed post-ingestion. Zero rows have NULL values in either movement column across the full 2,110,316-row dataset.

This was confirmed to mean Spotify used integer 0 for first-appearance rows rather than NULL. Stored procedures that need to detect a song's first appearance in a country must use MIN(snapshot_date) per song per country as the detection method, not NULL movement checks. This finding directly affected the design of sp_PopulateDiscoveryGapByDay.

9.3 Schema Changes Required During Ingestion

Table	Column / Constraint	Change	Reason
Album	name — NVARCHAR(255)	Widened to NVARCHAR(512)	Album name 'Rajieen (feat. Saif Safadi, Dana Salah, Ghaliaa, Afroto, NORDO...)' exceeded 255 characters
Album	UQ_Album_SpotifyID	Constraint dropped	All album_spotify_id values NULL pending API enrichment — UNIQUE prevents multiple NULLs

9.4 Cross-Dataset Summary

After both datasets were fully ingested, the combined database state is as follows:

Source	ChartEntry Rows	Date Range
Historical_Top200_and_Viral50	26,075,338	Jan 1 2017 – Dec 31 2021
Top50_73_Countries	2,110,316	Oct 18 2023 – Jun 11 2025
Total	28,185,654	Combined — 22-month gap confirmed